

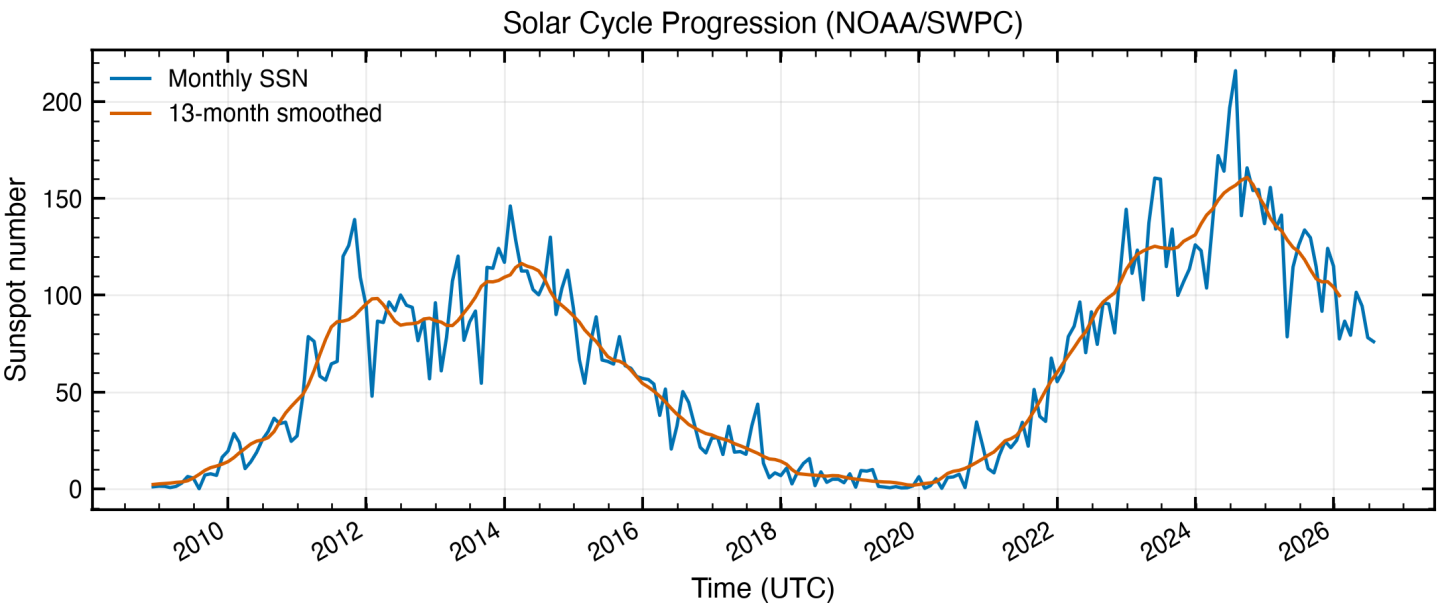
Reporting window and provenance

Window: 2026-09-01T11:00:00 to 2026-09-02T11:00:00 UTC. Sources: GOES primary XRS 1-min, NOAA RTSW solar wind at L1, NOAA planetary Kp, NOAA/SWPC solar region analysis, NASA CCMC DONKI, NOAA Solar Cycle Progression. Operational feeds - not science quality; every number is audit-logged (workspace/logs/audit.jsonl):

fetch_swpc_timeseries: ok (4c4c23f05648)
find_flares: ok (866403b2724c)
fetch_swpc_timeseries: ok (55fbe8afa20e)
fetch_swpc_timeseries: ok (17bec8351a6f)
fetch_swpc_timeseries: ok (b4025fab5741)
get_solar_regions: ok (f10bd36dcf41)
search_donki: ok (bc0f0672f58d)
fetch_solar_cycle: ok (c9f0a710f81e)
describe_series: ok (c287878f6dcb)
describe_series: ok (7d83aedcd365)
plot_timeseries: ok (9d0da3863f23)
plot_stack: ok (f2617d2edd4e)
plot_timeseries: ok (30170c5e421b)

Solar cycle

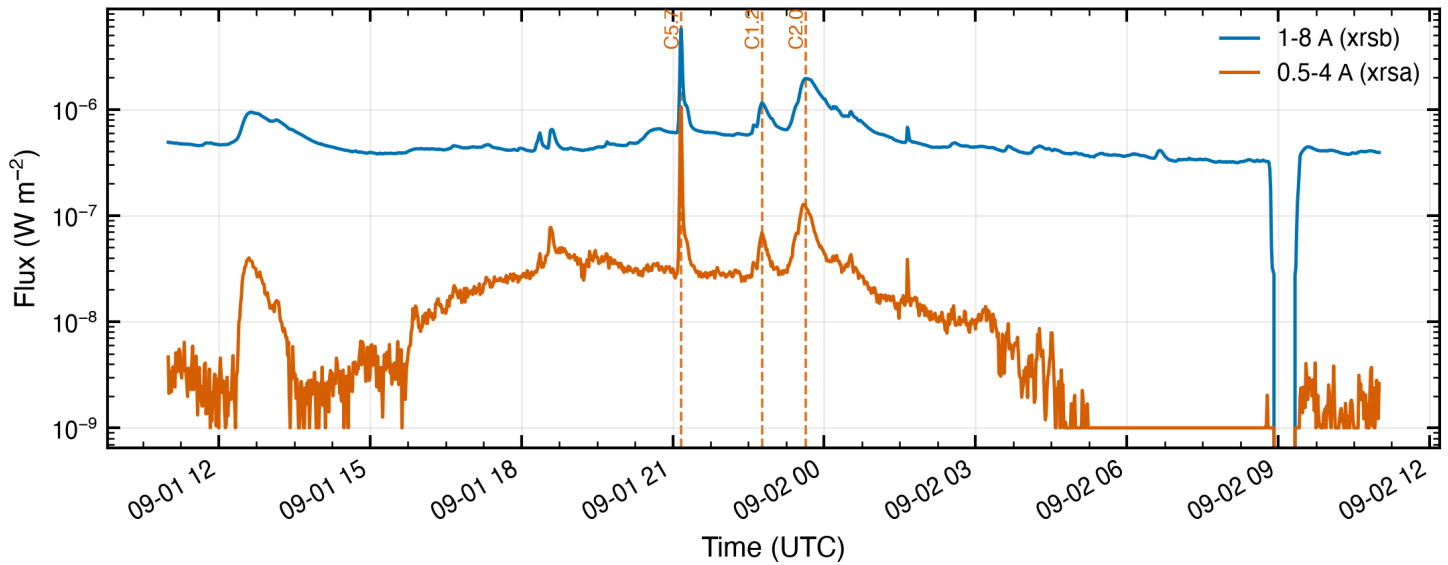
Latest monthly sunspot number: 76.0 (2026-08; previous month 78.1). Cycle 25 smoothed maximum 160.9 in 2024-10 (smoothed series final through 2026-02).



Flare activity

Detected 0 X, 0 M, 3 C flares (plus 33 B-level enhancements; sensitive detector - the official SWPC event list applies stricter criteria). Strongest: C5.7 (peak 21:10 UT, 118 min).

GOES XRS (operational), 2026-09-01 11:00 to 2026-09-02 11:00 UT



Sunspot regions (NOAA analysis)

Analysis date 2026-09-02:

AR4521 N09E39 mag A 1 spots, area 80 mu

AR4523 N10E49 mag A 2 spots, area 50 mu

AR4520 S12W33 mag B 4 spots, area 40 mu

CMEs (DONKI)

2026-09-01T06:00:00-CME-001: Faint CME observed to the SE in SOHO LASCO C2 and one frame of STEREO A COR2, fading into background and weakly visible in LASCO C3 and GOES CCOR-1. T

2026-09-01T21:24:00-CME-001: CME first seen to the East by SOHO LASCO C2 beginning at 2026-09-01T21:24Z, as well as by SOHO LASCO C3 and GOES CCOR-1 in later frames. This CME is n

2026-09-01T23:00:00-CME-001: CME first seen to the NE by SOHO LASCO C2 beginning at 2026-09-01T23:00Z, as well as by SOHO LASCO C3 and GOES CCOR-1 in later frames. This CME is not

2026-09-02T10:24:00-CME-001: Very faint CME first seen to the E in SOHO LASCO C2 starting around 2026-09-02T10:24Z. It is more easily seen to the E in STEREO A COR2. It is not see

Solar wind and geomagnetic conditions

Proton speed 360-609 km/s (mean 388); density mean 4.2 /cm³. Bt 2.4-6.5 nT; Bz GSM -4.5 to 5.5 nT. Kp 0.67-2.67.

Solar wind at L1 (RTSW) + Kp, 2026-09-01/2026-09-02

